LINEAR TRANSFORMATION

1 Topics

Linear transformations- definition, examples, basic properties. Problems: To check whether the given mapping is a linear transformation or not. Kernel and image of linear transformation. Proposition: Let $T : V \to U$ be a linear mapping. Then the kernel of T is a subspace of V and the image of T is a subspace of U. Show that a linear operator on a vector space V is one-one if and only if it is onto.

Prerequiste: vector space, basis, Functions, One-one and onto functions, Bijective functions.

Definition 1.1 *: Let V and W be vector spaces over a field F. A linear transformation from V to W (input to output) is a function such that T(cv+dw)=cT(v)+dT(w), for all v, w in V.*

Examples 1) Let V be a xy-plane. Rotate the plane by a fixed angle . Straight lines rotate to straight lines and origin stays put.
2) Derivative is a linear transformation
3) Also integral is a linear transformation
4) Reflection and Projection are linear transformation whereas translation is not.

Properties:

- (i)The definition of transformation implies that T(0)=0, i.e., T(0.v)=0.T(v), which implies T(0)=0, which is called zero transformation
- (ii) Linear Operator is a linear transformation from vector space to itself for example identity transformation is a mapping such that I(v)=v.
- (iii) Transformation preserves linear combination that is if $v_1, v_2, \cdots, v_n$ are some vectors in V and $c_1, c_2, \cdots, c_n$ are scalars, then $T(c_1v_1+c_2v_2+\cdots+c_nv_n) = c_1T(v_1) + c_2T(v_2) + \cdots + c_nT(v_n)$
- (iv) T(-v)=-T(v)

- (v)T(v-w)=T(v)-T(w)
- (vi) Composition of linear Transformation is a linear mapping
- (vii) A function is said to be invertible if there exists a function U from W into V such that UT is identity function on V and TU is identity function on W. We denote $U = T^{-1}$ and is also a linear transformation.

Problems:

Example 1 *Consider* $T : R^3 \to R^3$ *defined by* $T(x_1, x_2, x_3) = (x_1, x_2, 0)$, *for all* $(x_1, x_2, x_3) \in R^3$. *Check if* T *is a linear transformation.*

Solution:

$$\begin{aligned} T((x_1, x_2, x_3) + (y_1, y_2, y_3) &= T(x_1 + y_1, x_2 + y_2, x_3 + y_3) \\ &= (x_1 + y_1, x_2 + y_2, 0) \\ &= (x_1, x_2, 0) + (y_1, y_2, 0) \\ &= T(x_1, x_2, x_3) + T(y_1, y_2, y_3). \end{aligned} \tag{1.2}$$

and

$$\begin{aligned} T(\alpha(x_1, x_2, x_3) &= T(\alpha x_1, \alpha x_2, \alpha x_3) \\ &= (\alpha x_1, \alpha x_2, 0) \\ &= \alpha(x_1, x_2, 0) \\ r &= \alpha T(x_1, x_2, x_3). \end{aligned} \tag{1.3}$$

From (1.4) and (1.3), T is linear transformation..

Similarly,

Example 2 *Let* $T : R^2 \to R^2$ *defined by* $T(x, y) = \begin{pmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$, *where* θ *ia fixed angle.*

Example 3 *Let* $T : R^2 \to R^3$ *defined by* $T(x, y) = (x + 3, 2y, x + y)$ *is not a linear transformation.*

Solution:

$$\begin{aligned} T((x_1, x_2) + (y_1, y_2)) &= T(x_1 + y_1, x_2 + y_2) \\ &= (x_1 + y_1 + 3, 2(x_2 + y_2), x_1 + y_1 + x_2 + y_2) \\ &\neq (x_1 + 3, 2y_1, x_1 + y_1) + (x_2 + 3, 2y_2, x_2 + y_2) \\ &\neq T(x_1, x_2) + T(y_1, y_2). \end{aligned} \tag{1.4}$$

Therefore, T is not linear.

2 Kernel and Image of Linear Transformations

Definition 2.1 *If T is a linear transformation from V into W, then range of T is te set of all vectors $w \in W$ such that $w = Tv$ for some $v \in V$. The dimension of image space or range of T is called rank of T.*

Definition 2.2 *If T is a linear transformation from V into W, then the set of all vectors $v \in V$ such that $Tv = 0$ is called kernel of T or null space of The dimension of ker T is called nullity.*

Proposition 2.1 *Let $T : V \to W$ be a linear mapping. Then kernel of T is a subspace of V.*

Proof: To show that $kerT = \{v/Tv = 0\}$ is closed wrt addition and scalar multiplication and also $0 \in kerT$.

Since T is linear, T(0) =0 which implies $0 \in kerT$. Let $u, v \in kerT$, then $T(u) = 0$ and $T(v) = 0$. Since u and v are in V, which is a vector space, v+u also lies in V. Hence consider T(u+v). By linear transformation,

$$\begin{aligned} T(u+v) &= T(u) + T(v) \\ &= 0 \\ &\implies u + v \in kerT \end{aligned} \tag{2.3}$$

Similarly, $T(\alpha u) \in kerT$.

Proposition 2.2 *Let $T : V \to W$ be a linear mapping. Then image of T is a subspace of W.*

Proof: To show that range T is closed wrt addition and scalar multiplication and also $0 \in rangeT$.

Since T is linear, T(0) =0 which implies $0 \in Img\, T$. Let $w_1, w_2 \in Img\, T$, then there exists $v_1, v_2 \in V$ such $T(v_1) = w_1$ and $T(v_2) = w_2$. Since v_1, v_2 are in V, which is a vector space. Hence, consider $T(v_1 + v_2)$. By linear transformation,

$$\begin{aligned} T(v_1 + v_2) &= T(v_1) + T(v_2) \\ &= w_1 + w_2 \\ &\implies w_1 + w_2 \in Img\, T \end{aligned} \tag{2.4}$$

Similarly, $T(\alpha u) \in Img\, T$.

Proposition 2.3 *Suppose $v_1, v_2, \cdots, v_n$ spans a vector space V and $T : V \to W$ is linear. Then $T(v_1), T(v_2), \cdots, T(v_n)$ spans Img T.*

Proof: let $v_1, v_2, \cdots, v_n$ spans a vector space V and $T : V \to W$ be linear. Let $w \in ImgT$. Then there exists $v \in V$ such that $T(v) = w$. Since, $\{v_i\}$'s span V and $v \in V$, there exists scalars $a_1, a_2, \cdots, a_n$ for which

$$\begin{aligned} v &= a_1v_1 + a_2v_2 + \cdots + a_nv_n \\ T(v) &= T(a_1v_1 + a_2v_2 + \cdots + a_nv_n) \\ w &= a_1T(v_1) + a_2T(v_2) + \cdots + a_nT(v_n) \\ &\Longrightarrow \{T(v_1), T(v_2), \cdots, T(v_n)\} \textit{spans the image of } F. \end{aligned}$$

Note: If T is non singular, nullity of T is zero. This implies that T is onto.

Lemma 2.1 *Let $T : V \to W$ be linear map. Then T is injective iff $null(T) = \{0\}$.*

Proof Let T is one one that is $Tv = Tw \Rightarrow v = w$. Since null(T) is a subspace of V, $0 \in null(T)$. Let $v \in kerT$. This implies that $T(v) = 0$, by linearity, $T(v) = T(0)$. Since T is injective, v=0. Therefore $0 \in kerT$. Hence, $null(T) = \{0\}$.

Conversely, let $null(T) = \{0\}$. Let $u, v \in V$ such that $Tu = Tv, \Rightarrow T(u - v) = 0$, $u - v \in null(T) = \{0\}$. Therefore, u=v.

Lemma 2.2 *Let $T : V \to W$ be linear map. If T is onto then $img(T) = W$.*

Proof is obvious by Rank Nullity theorem.

Proposition 2.4 *A linear operator ona vector space is one one iff it is onto.*

Proof: Let $T : V \to W$ be linear and dim V=dim W=n. Let T be one one then $null(T) = \{0\}$. Hence, by rank-nullity theorem, dim (Img T)=n, which is dim V. This implies that $T(V) = W$. Therefore, T is onto. Conversely, if T is onto, $T(V) = W$, range of T is W, which implies that rank T is n. Then by rank nullity theorem, nullity is zero, which implies that $null(T) = \{0\}$ and hence, T is one one.

Example 4 *Let $T : R^2 \to R^3$ be such that T(x,y)=(x+y,x-y,x), then check if T is one one and T is onto?*

Solution: kerT=$\{(0, 0)\}$. So T is on one. Img T=$\{x(1, 0, 1/2) + y(0, 1, 1/2)\}$ which implies that dim(Img T)=2 which is not dim R^3. So, T is not onto.